05. REVISIONS : CHRONOLOGY OF SYSTEMS, THE 5 PHASES OF EMBRYOLOGY

DURATION : 24:00

STUDY SHEET

COURSE SUMMARY

This video addresses the chronology of communication systems in embryonic development, focusing on five key phases: fertilisation, gastrulation, neurulation, metamerisation, and delimitation. You will learn how the digestive and circulatory systems develop in interaction, as well as the fundamental role of the vitelline cavity and the blastocoel in embryonic organisation. Basic principles such as axial formation and tissue dynamics are explained, illustrating how these processes influence the health and integration of body systems in a biodynamic approach. This knowledge is essential for therapists wishing to deepen their understanding of the embryological origins of human structures and their application in osteopathy.

CHRONOLOGY OF SYSTEMS: THE 5 PHASES OF EMBRYOLOGY

In the **chronology of the appearance of communication systems**, the **digestive system** is very early. The first movement in this system is a fluidic, **autocrine** type of movement. In other words, it's about learning to know oneself before being able to exchange and digest.

The **conjunctive circulatory endocrine system** appears next. The endocrine system depends on the circulatory system. First, the **enteric system** develops, followed by the **parasympathetic system** and then the **orthosympathetic system**. This means that the first brain is digestive; the enteric precedes cerebralisation.

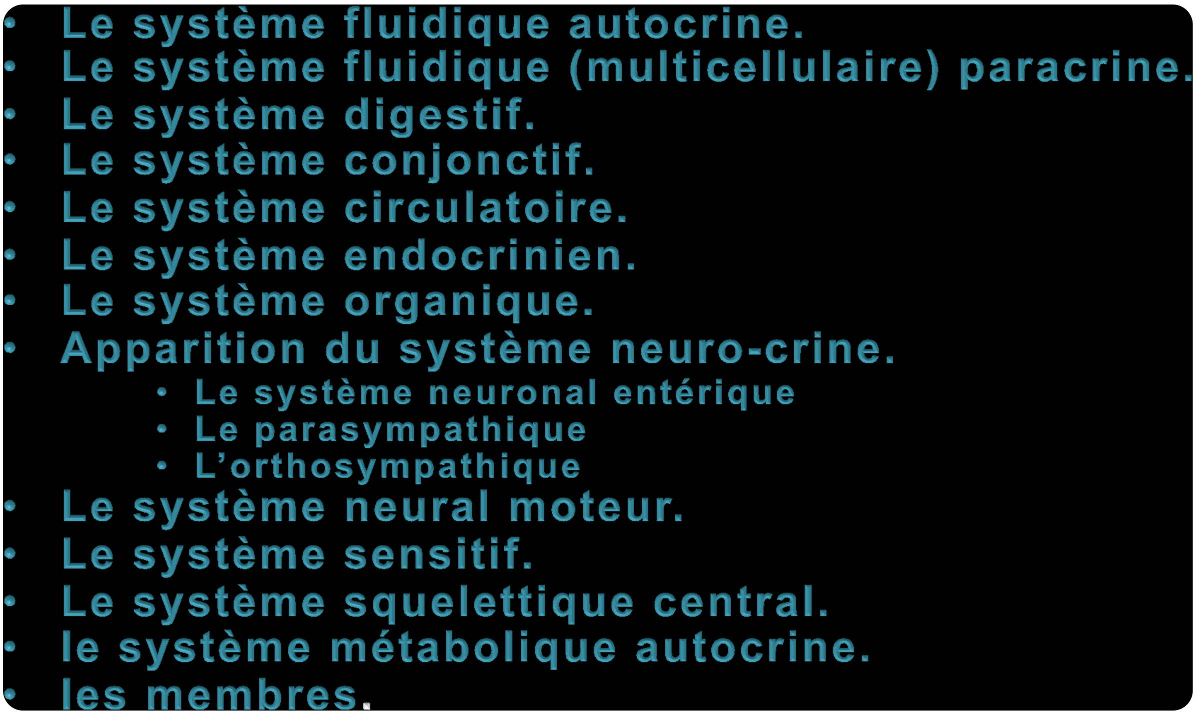

FIG. — Listed body systems

Do not forget the five major phases of embryology: **fertilisation, gastrulation, neurulation, metamerisation, and delimitation**. On the first day, an axis is established between the two polar bodies, essential for digestive development. Each cell division or cleavage leads to an increase in the **metabolic system** and the **cell membrane**, but also a small loss of water.

This continuous movement during development causes the appearance of a cavity called the **blastocoel**. This cavity represents the anlage of the **vitelline cavity**, which will be integrated into the digestive tract. It is important to note that this cavity is always off-centre, responding to global polarity.

The embryo begins to organise itself with a structure of **micromeres**: a vegetal pole, a swollen pole, and an assimilating pole, representing a force to seek trophic information. The **blastocoel** is a cavity open to the outside that gradually integrates inwards, which means the digestive tract represents a quality called **humus**. This humus is comparable to the **intestinal flora**, symbolising the living quality of the intestine, nourished from the outside.

The establishment of the **vitelline cavity** and the **external coelom** occurs in the **uterine mucosa**. The **trophoblast**, a black layer, is a blastocoel surrounded by fluid, allowing development between the fluid layers. The cerebral ectodermal tissue is in relation to the amniotic cavity, while the tissue facing the primitive coelomic cavity will become the digestive system.

This growth process is organised by fulcrums and by differential growth rates. The peripheral layer, in contact with the uterus, receives nutritional information. If feeding is rapid, so is growth. However, the rapidly growing peripheral layer creates a tearing between it and the central layer, forming **Heuser's membrane** which maintains the cavity.

This tearing transforms the blastocoel into the vitelline cavity and external coelom. At this stage, a primitive embryonic pellicle organises itself, already present during the integration of the primitive embryo at implantation. This phase organises an S-shape within the embryonic disc, with a fulcrum between an expansion dome and an impaction dome, essential for the adaptive process.

The axial formation of the embryo develops, leading to the **notochordal** phase. This S-shape is influenced by fields of **permeation**, **infusion**, and **parmeation**, organising the amniotic vesicle anteriorly and the vitelline vesicle posteriorly. Cells in the convergent part grow faster than those in the convex part, with a fulcrum called the **zero point**, the anlage of the primitive craniosacral axis.

This axial process is fundamental for the overall development of the embryo, serving as an organising centre. It is linked to the base of the skull and the notochordal axis down to the sacrum, developing between the umbilicus and the base of the skull. The heart, diaphragm, and liver form in a dynamic of invagination and evagination, modifying the fulcrum.

The rupture of symmetry leads to a nodal flow that changes the molecular aspect of the embryo, influencing genomic organisation. The body knows how to reorganise itself, first at the navel, then the sphenobasilar, the heart, the diaphragm, the liver, the adrenals, and the gonads. The sacrum and the base of the skull form a functional unit.

Embryonic development is characterised by an anterior movement of the cavity and a posterior development of the vitelline vesicle. Cells facing the anterior dome develop faster than those at the posterior, creating a point of overall non-growth. The future base of the skull, at the tip of the notochord, is surrounded by para-axial and hypophyseal cartilage, which will become the occipital and sphenoidal system.

Embryonic flexion and fluid dynamics are essential for understanding coiling and growth. On day 28, the integration of the external coelom into an internal coelom marks a fulcrum on the cerebrospinal fluid. Development coils around the vascular body, requiring fluids to be brought back to the centre.

It is crucial to listen to the movement of the tissue and find its balance point to enter its history. This process can reveal images or words related to embryonic points or transgenerational capacities.

FIG. — 2-cell embryo

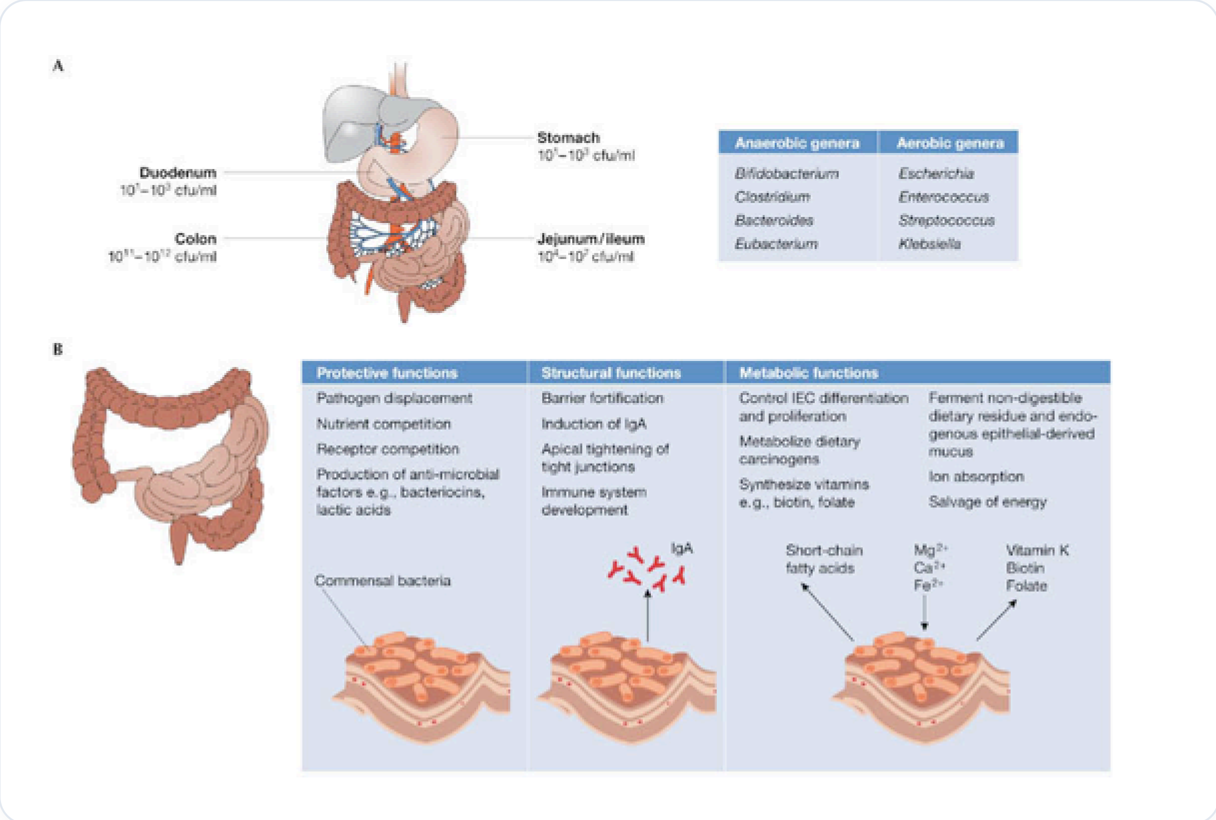

FIG. — Gut microbiota functions

FIG. — Intestinal flora microbiota